IND3001 Engineering Economics

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Problem-solving process

- 1) Problem definition / Determination of project objectives
- 2) Data collection
- 3) Identification of alternative solutions
- 4) Determination of cash flows and outcomes (for all alternatives)
- 5) Selection of evaluation criteria

- AW (Annual Worth)

- ROR (Rate of Return)

- C/B (Cost-Benefit Analysis)

- PP (Payback Period)

- 6) Engineering economy analyses (considering non-economic factors, sensitivity, and risk analyses)
- 7) Selection of the best alternative
- 8) Implementation and monitoring

Fundamental principles of engineering economics

- 1) Money in hand is more valuable than money to come.
- 2) Preferences are formed based on the differences between alternatives.
- 3) Marginal revenue must be greater than marginal cost. MR>MC

  Marginal revenue (MR) is the additional revenue obtained from increasing production by one unit. Marginal cost (MC) is the additional cost incurred for a one-unit increase in production.
- If MR > MC, production continues; if MR < MC, production does not continue.
- 4) Additional risk should not be taken without the expectation of additional income.

Types of Cost

- 1) Investment Cost:

It is used to express the total costs incurred when initiating an activity. These costs generally occur once and do not recur.

2) Fixed Cost (FC), Variable Cost (VC) and Mixed Cost

FC: These are costs that remain unaffected by changes in production volume. They are costs that do not vary with the level of production, such as general administrative expenses, taxes, insurance and license fees, management personnel costs, and depreciation of buildings and machinery.

As the production volume increases, the fixed cost per unit decreases; as the production volume decreases, the fixed cost per unit increases.

VC:Variable costs fluctuate depending on the quantity of production. For instance, when production volume increases by 5%, total variable costs will also increase by approximately 5%.

Labor and materials used to produce goods and services constitute variable costs.

Types of Cost

- 3) Total, Average and Marginal Cost:

TC: Total cost is the sum of fixed and variable costs.

$TC(x) = FC + VC(x)$ x = annual machine hours.

AC: Average cost is the average cost per unit of production, representing the ratio of total cost to the quantity of production.

$AC(x) = TC(x) / x$

MC:Marginal cost is the increase in total cost resulting from a one-unit increase in production. In other words, it is the rate of change in total cost in response to a change in the quantity of production.

$MC = dTC / dx$

Types of Cost

- 4) Sunk Cost:

A sunk cost is a cost that has already been incurred in the past and is irrelevant to the cost and revenue estimates for any alternative course of action; in other words, it is common to all alternatives.

In engineering economy analyses, it can be disregarded.

A sunk cost is money that has been spent and is gone; no current or future activity can recover it.

5) Opportunity Cost:

Opportunity cost is the loss of the chance to earn income from an alternative investment when a specific investment is made. It can be thought of as the cost of the best alternative opportunity that is forgone.

Opportunity Cost

- If an individual uses $30,000 in savings to purchase stocks, they estimate that they will earn an annual income of approximately $3,000.
- If they choose to buy a car instead of stocks, they will sacrifice that $3,000 annual income.
- This $3,000 represents the annual opportunity cost of purchasing the car.

Economic analyses of the current situation

- When the alternatives being compared relate to a specific situation, cover a period of one year or less, and the time value of money can be neglected, the engineering economy analyses are referred to as economic analyses of the current situation.

- There are two rules to keep in mind for these analyses:
- 1) When there are revenues or other economic benefits that vary across alternatives, the alternative that maximizes total profitability should be chosen.
- 2) When there are no revenues or other economic benefits, or when they are the same for all alternatives, only costs should be considered, and the alternative that minimizes total cost should be chosen.

Choosing among alternatives

1. Case: A situation in which the relationships between interest, money, and time do not exist

   When choosing between different materials, one should not consider only the price of the material; factors such as the type of material and transportation costs—which can also affect overall expenses—must also be taken into account.

2. Case: The problem of location that is independent of time

3. Case: Time-independent efficiency and production cost

   Employees with different levels of productivity produce different quantities of goods/services in a given amount of time.

4. Case: Choosing the most economical production machine
5. Case: Choosing the appropriate pump

Thanks!
Q&A

